1 Hilbert Proofs

1.1 Formulas, axioms, and proofs

Definition 1.1 (formulas)**.** Fix a stock of letters $p, q, r, \ldots$. The formulas are defined inductively:

1. Every letter is a formula
2. If φ and ψ are formulas, so is $\varphi \to \psi$, read "if φ, then ψ.
3. Nothing else is a formula

In one line:

$$\varphi, \psi \quad := \quad p \quad | \quad \varphi \to \psi$$

The arrows groups to the right:

$$\varphi \to \psi \to \chi \text{ is } \varphi \to (\psi \to \chi)$$

Small Greek letters stand for formulas, as capital Latin ones stood for types. Formulas have the same grammar as types; Section 2.2 says why.

What do we mean by "if ..., then ...?" At least this: whenever the *if* parts holds, the *then* part may be concluded. In symbols,

$$\text{from } \varphi \to \psi \text{ and } \varphi, \text{ conclude } \psi$$

The rule's Latin name is *modus ponens*, and it is the only move we make in this chapter. The first shape: A truth is implied by anything: if φ holds, then φ holds whatever else, say ψ, does.

$$\varphi \to \psi \to \varphi$$

The second shape: *modus ponens* carried out under a hypothesis: if, given φ, we have $\psi \to \chi$, and given φ, we have ψ, then given φ, we have χ.

$$(\varphi \to \psi \to \chi) \to (\varphi \to \psi) \to \varphi \to \chi$$

Definition 1.2 (proof from hypotheses)**.** The *axioms* are all instances of two schemas, where φ, χ and χ stand for arbitrary formulas:

$$\begin{array}{ll} \text{(I)} & \varphi \to \psi \to \varphi \\ \text{(II)} & (\varphi \to \psi \to \chi) \to (\varphi \to \psi) \to \varphi \to \chi \end{array}$$

Let Γ be a set of formulas, the hypotheses. A proof of φ from Γ is a finite sequence of formulas $\varphi_1, \ldots, \varphi_n$ whose last line is φ, in which each line φ_k is justified in one of three ways:

1. (*axiom*) ϕ_k is an instance of (I) or (II);
2. (*hypothesis*) is a member of Γ;
3. (*modus ponens*) there are earlier lines φ_i and φ_j, with $i, j < k$ such that φ_i is the formula $\varphi_j \to \varphi_k$.

No other kind of line is allowed.

Definition 1.3 (theorems)**.** If such a proof exists, we write

$$\Gamma \vdash \varphi$$

read "φ is provable from Γ". A proof from Γ does not know whether its hypotheses hold; it says what follows if they do. When Γ is empty the second clause never applies. We write $\vdash \varphi$ and call φ a theorem: it holds by virtue of its logical structure alone, with nothing told.

Example 1.4 (is it an axiom?)**.** Is each an instance of (I), (II), or neither?

1. $p \to q \to r$: Neither. (I) needs the same formula at both ends.
2. $q \to (p \to p) \to q$: (I) with $\varphi := q$, $\psi := p \to p$
3. $(p \to q) \to (q \to r) \to p \to r$ Neither. It is what chaining two implications concludes, not an instance of a schema.
4. $(p \to p \to p) \to (p \to p) \to p \to p$ (II) with $\varphi := p$, $\psi := p$, $\chi := p$

Checking a shape is a matter of comparing the pieces, no understanding required.

Example 1.5 ($\vdash p \to p$)**.**

	no hypotheses	
1	$(p \to (p \to p) \to p) \to (p \to (p \to p)) \to p \to p$	(II) $[p,\ p \to p,\ p]$
2	$p \to (p \to p) \to p$	(I) $[p,\ p \to p]$
3	$(p \to (p \to p)) \to p \to p$	MP 1, 2
4	$p \to (p \to p)$	(I) $[p,\ p]$
5	$p \to p$	MP 3, 4

Each line is trivial to check, but Line 1 is not trivial to find. The thinking runs backwards from the goal. That is how all proofs are found.

1. The goal $p \to p$ is not an axiom, so it comes by modus ponens: from some χ and $\chi \to (p \to p)$.
2. Could $\chi \to (p \to p)$ be an instance of (I)? Then χ would be p, and we would need a proof of p on its own. There cannot be one: nothing in the system proves a bare letter.
3. So try (II). Its last part, $\varphi \to \chi$, must be $p \to p$. That fixes $\varphi = \chi = p$ and leaves ψ free.
4. The two premises are then $p \to (\psi \to p)$ and $p \to \psi$. Choose $\psi := p \to p$, and both are instances of (I).

1.2 Proofs are programs

Theorem 1.6 (Curry's correspondence)**.** $\Gamma \vdash M : A$ for some term M if and only if $|\Gamma| \vdash A$

1. (Erasure) Replace every line of a typing of $\Gamma \vdash M : A$ by its partner in the right-hand column. The result is a proof of A from $|\Gamma|$
2. (Decoration) Take a proof of φ from some hypotheses, give each hypothesis a variable, and replace every line by its partner in the left-hand column, working downwards. The result is a typing of $\Gamma \vdash M : \varphi$ for some term M.

All that needs checking is that each line's justification survives the change. Every line of a typing has the same context Γ, so take the four kinds of line in turn.

1. A line by (K) has type $B \to C \to B$: erased, an instance of (I).
2. A line by (S) has type $(B \to C \to D) \to (B \to C) \to B \to D$: erased, an instance of (II)
3. A line by (var) has a type listed in Γ: erased, a member of $|\Gamma|$, a hypothesis.
4. A line by "(app) i, j" has type C, where line i has type $B \to C$ and line j has type B. Erased, "MP i, j" is a correct justification for C.

It does not say that proofs *resemble* programs, or that there is a useful analogy.

1.3 More connectives: and, or, and not

Definition 1.7 (formulas, extended)**.** Fix a stock of letters $p, q, r, \ldots$, as before. The formulas are defined inductively:

1. every letter is a formula;
2. $\bot$ is a formula, read "absurd" or "that cannot happen"
3. if φ, ψ are formulas, so is $\varphi \to \psi$;
4. if φ, ψ are formulas, so is $\varphi \wedge \psi$, "φ and ψ";
5. if φ, ψ are formulas, so is $\varphi \vee \psi$, "φ or ψ";
6. nothing else is a formula.

In one line:

$$\varphi, \psi \quad ::= \quad p \quad | \quad \bot \quad | \quad \varphi \to \psi \quad | \quad \varphi \wedge \psi \quad | \quad \varphi \vee \psi$$

$\neg\varphi$ abbreviates $\varphi \to \bot$, read "not φ": a short way of of writing an arrow into $\bot$.

Definition 1.8 (nine schemas)**.** The axioms are all instances of nine schemas, where φ, ψ, χ stand for arbitrary formulas.

SCHEMA	SHAPE
(I)	$\varphi \to \psi \to \varphi$
(II)	$(\varphi \to \psi \to \chi) \to (\varphi \to \psi) \to \varphi \to \chi$
(pair)	$\varphi \to \psi \to \varphi \wedge \psi$
(fst), (snd)	$\varphi \wedge \psi \to \varphi, \quad \varphi \wedge \psi \to \psi$
(left), (right)	$\varphi \to \varphi \vee \psi, \quad \psi \to \varphi \vee \psi$
(either)	$(\varphi \to \chi) \to (\psi \to \chi) \to \varphi \vee \psi \to \chi$
(absurd)	$\bot \to \varphi$

The same three kind of line: a proof of φ from Γ is a box, as before, in which each line is

1. (*axiom*) an instance of one of the nine schemas;
2. (*hypothesis*) a member of Γ;
3. (*modus ponens*) the φ_k of earlier lines $\varphi_j \to \varphi_k$ and φ_j.

AND: With r "Smith may publish", a "Smith may edit the release notes".

SCHEMA	SHAPE	HASKELL	FOR HUGHES
(pair)	$\varphi \to \psi \to \varphi \wedge \psi$	`(,)`	if Smith may publish, then, if he may edit the notes, he may publish and edit the notes
(fst)	$\varphi \wedge \psi \to \varphi$	`fst`	someone who may publish and edit the notes may publish
(snd)	$\varphi \wedge \psi \to \psi$	`snd`	and may edit the notes

To make a conjunction, you need both halves. To use one, you take a half.

OR: With b "the scheduled job triggered the release", h "a maintainer pressed the button", r "the release may go ahead".

SCHEMA	SHAPE	HASKELL	FOR HUGHES
(left)	$\varphi \to \varphi \vee \psi$	`Left`	if the job triggered it, the job or a maintainer did
(right)	$\psi \to \varphi \vee \psi$	`Right`	likewise for the maintainer
(either)	$(\varphi \to \chi) \to (\psi \to \chi) \to \varphi \vee \psi \to \chi$	`either`	if the job may publish, and a maintainer may, then whichever did it, it may go ahead

To make a disjunction you need one side. To use one you must handle both: (either) is proof by cases.

NOT: A new formula $\bot$, read "absurd": *that cannot happen.* Nothing makes $\bot$.

$$\text{(absurd)} \quad \bot \to \varphi$$

From what cannot happen, anything follows. Haskell: `absurd :: Void -> a`. And $\neg\varphi$ abbreviates $\varphi \to \bot$, read "not φ". "A fork may not publish" is $u \to \neg r$, that is, $u \to r \to \bot$. u: "the request comes from a fork". A prohibition is an arrow into absurdity.

Example 1.9 (proof by cases)**.** The bot may publish when the job triggers it ($b \to r$), and when a maintainer presses the button ($h \to r$). The log says one of the two happened ($b \vee h$), not which. May the release go ahead?

	$b \to r, h \to r, b \vee h$	
1	$b \to r$	hypothesis
2	$h \to r$	hypothesis
3	$(b \to r) \to (h \to r) \to b \vee h \to r$	(either) $[b, h, r]$
4	$(h \to r) \to b \vee h \to r$	MP 3, 1
5	$b \vee h \to r$	MP 4, 2
6	$b \vee h$	hypothesis
7	r	MP 5, 6

Two rules with the same conclusion are as good as one rule whose condition is "the one or the other". That is what (either says).

Example 1.10 (commutativity of $\vee$)**.**

	no hypotheses	
1	$(b \to \varepsilon) \to (h \to \varepsilon) \to b \vee h \to \varepsilon$	(either) $[b, h, \varepsilon]$
2	$b \to \varepsilon$	(right) $[h, b]$
3	$(h \to \varepsilon) \to b \vee h \to \varepsilon$	MP 1, 2
4	$h \to \varepsilon$	(left) $[h, b]$
5	$b \vee h \to \varepsilon$	MP 3, 4

No hypotheses and no discharge: the two cases were both axioms, and (either) joined them.

Example 1.11 (from a contradiction, anything)**.** A careless edit leaves the policy saying both that Smith has write access and that he does not. What follows? Everything: any r, "Jones may rotate the signing key", say.

	$q, \neg q$	
1	q	hypothesis
2	$q \to \bot$	hypothesis (that is, $\neg q$)
3	$\bot$	MP 2, 1
4	$\bot \to r$	(absurd) $[r]$
5	r	MP 4, 3

"Not q" means that q leads to what cannot happen; but q holds; so what cannot happen has happened; and from that, anything. A conflict between an allow rule and a deny rule is an emergency.

Example 1.12 (commutativity of $\wedge$)**.** "May publish and edit notes" written by the colleague the other way around. r "may publish", a "may edit the notes". For now: told both, conclude both the other way round.

	$r \wedge a$	
1	$r \wedge a$	hypothesis
2	$r \wedge a \to a$	(snd) $[r, a]$
3	a	MP 2, 1
4	$r \wedge a \to r$	(fst) $[r, a]$
5	r	MP 4, 1
6	$a \to r \to a \wedge r$	(pair) $[a, r]$
7	$r \to a \wedge r$	MP 6, 3
8	$a \wedge r$	MP 7, 3

Example 1.13 (contraposition)**.** The checker refuses Smith a release: he has no write access. That tells Hughes he is not a maintainer. She wants $\vdash (p \to q) \to \neg q \to \neg p$. The clerk can follow her as far as "which cannot happen".

	$p \to q, \neg q, p$	
1	$p \to q$	hypothesis
2	$q \to \bot$	hypothesis (that is, $\neg q$)
3	p	hypothesis
4	q	MP 1, 3
5	$\bot$	MP 2, 4

$\{p \to q, \neg q, p\} \vdash \bot$. That is what $\neg p$ means, but the clerk has p above the line, where it is of no use to anyone.

1.4 Proof theory

Proposition 1.14 (structural facts)**.** For all sets of Γ, Δ and all formulas φ, ψ:

1. (reflexivity) if $\varphi \in \Gamma$ then $\Gamma \vdash \varphi$;
2. (monotonicity) if $\Gamma \subseteq \Delta$ and $\Gamma \vdash \varphi$ then $\Delta \vdash \varphi$;
3. (transitivity) if $\Gamma \vdash \varphi$ and $\Delta \cup \{\varphi\}$ then $\Gamma \cup \Delta \vdash \varphi$.

1. The one-line sequence φ is a proof of φ from Γ: its only line is justified as a hypothesis, because φ is a member of Γ.
2. A proof of φ from Γ is already a proof of φ from Δ: every line justified as a member of Γ is a member of Δ too, and the two kinds of line do not mention Γ.
3. Write out the proof of φ from Γ, and after it the proof of ψ from $\Delta \cup \{\varphi\}$ with every line justified as "hypothesis φ" deleted; each modus ponens that cited one of the deleted lines now cites the last line of the first proof, which is φ. Every remaining hypothesis is in Γ or in Δ, so the result is a proof of ψ from $\Gamma \cup \Delta$.

Example 1.15 (a fact once told stays told)**.** Show $\{p\} \vdash q \to p$: from "Smith is a maintainer", "if he has write access, he is a maintainer".

	p	
1	$p \to q \to p$	(I) $[p, q]$
2	p	hypothesis
3	$q \to p$	MP 1, 2

Example 1.16 (hypothetical syllogism)**.** The rule that chains two implications. From the two sentences of the policy it derives the third that nobdoy wrote down: if Smith is a maintainer, he may publish. Show $\{p \to q, q \to r\} \vdash p \to r$.

	$p \to q, q \to r$	
1	$q \to r$	hypothesis
2	$(q \to r) \to p \to (q \to r)$	(I) $[q \to r, p]$
3	$p \to (q \to r)$	MP 2, 1
4	$(p \to (q \to r)) \to (p \to q) \to p \to r$	(II) $[p, q, r]$
5	$(p \to q) \to p \to r$	MP 4, 3
6	$p \to q$	hypothesis
7	$p \to r$	MP 5, 6

Read the block, not the symbols:

- Lines 1–3: Take the hypothesis $q \to r$ and weakn it, by (I), to $p \to (q \to r)$: a statement that says less, since it only promises $q \to r$ once p is given.
- Lines 4–5: Put that through (II), which turns "given p, q implies r" into "if p gives q then p gives r".
- Line 7: Fire the result on the other hyptohesis.

Example 1.17 (the filled rule)**.** By transitivity certain lines never have to be written.

	$p, p \to q, q \to r$	
1	$p \to r$	hypothetical syllogism
2	p	hypothesis
3	r	MP 1,2

Line 1 is not an axiom and not a hypothesis. It is justified by a fact about proofs.

Example 1.18 (chaining three)**.** Or chain three implications instead of two: $\{p \to q, q \to r, r \to s\} \vdash p \to s$ is two uses of the filled rule.

	no hypotheses	
$p \to q, q \to r, r \to s$ 1	$p \to r$	hypothetical syllogism
2	$p \to s$	hypothetical syllogism, from 1 and $r \to s$

1.5 The Deduction Theorem

Theorem 1.19 (Deduction Theoreom)**.** For every set of formulas Γ and all formulas φ, ψ:

$$\Gamma \cup \{\varphi\} \vdash \psi \iff \Gamma \vdash \varphi \to \psi$$

Right to left is one use of modus ponens:
Suppose $\Gamma \vdash \varphi \to \psi$. Show $\Gamma \cup \{\varphi\} \vdash \psi$.

1. Take proof of $\varphi \to \psi$ from Γ. It is also a proof from $\Gamma cup\{\varphi\}$, by monotonicity.
2. Add the line φ, justified as a hypothesis: φ is a member of $\Gamma \cup \{\varphi\}$.
3. Add the line ψ, by modus ponens on $\varphi \to \psi$ and φ.

This is the easy direction: we can make use of the implication.

Left to right is the transformation. The plan:

- Let $\varphi_1, \ldots, \varphi_n = \psi$ be a proof of ψ from $\Gamma \cup \{\varphi\}$.
- Go down it one line at a time.
- For each line φ_k, write out a proof of $\varphi \to \varphi_k$ from Γ *alone*, using the proofs already writen for the earlier lines.
- At the last line, $\varphi_n = \psi$, we have a proof of $\varphi \to \psi$.

Induction on k: **Hypothesis**: for every $j < k$, a proof of $\varphi \to \varphi_j$ from Γ has been written.
To show: a proof of $\varphi \to \varphi_k$ from Γ, built on top of those. Three cases:

1. φ_k is an axiom or a member of Γ:
   Then, φ_k is a legitimate line of a proof from Γ. Write:

	Γ	
1	φ_k	as before
2	$\varphi_k \to \varphi \to \varphi_k$	(I) $[\varphi_k, \varphi]$
3	$\varphi \to \varphi_k$	MP 2, 1

   Three lines: A line we did not need φ for stays true when we suppose φ as well, (I) says exactly that.

2. φ_k is φ itself:
   Then $\varphi \to \varphi_k$ is $\varphi \to \varphi$, and the five-line proof earlier, with φ for p, proves it from no hypotheses at all, so in particular from Γ.

	Γ	
1	$(\varphi \to (\varphi \to \varphi) \to \varphi) \to (\varphi \to (\varphi \to \varphi)) \to \varphi \to \varphi$	(II) $[\varphi, \varphi \to \varphi, \varphi]$
2	$\varphi \to (\varphi \to \varphi) \to \varphi$	(I) $[\varphi, \varphi \to \varphi]$
3	$(\varphi \to (\varphi \to \varphi)) \to \varphi \to \varphi$	MP 1, 2
4	$\varphi \to (\varphi \to \varphi)$	(I) $[\varphi, \varphi]$
5	$\varphi \to \varphi$	MP 3, 4

3. modus ponens:
   Then, there are earlier lines φ_j and $\varphi_i = \varphi_j \to \varphi_k$. By the induction hypothesis, proofs from Γ of $\varphi \to \varphi_j$ and of $\varphi \to \varphi_j \to \varphi_k$ have already been written.

	Γ	
1	$(\varphi \to \varphi_j \to \varphi_k) \to (\varphi \to \varphi_j) \to \varphi \to \varphi_k$	(II) $[\varphi, \varphi_j, \varphi_k]$
2	$(\varphi \to \varphi_j) \to \varphi \to \varphi_k$	MP a, $(\varphi \to \varphi_j \to \varphi_k)$
3	$\varphi \to \varphi_k$	MP b, $(\varphi \to \varphi_j)$

Three lines, on top of the proofs already written. Only this case uses the IH. Concatenating the pieces in order gives a proof from Γ whose last line is $\varphi \to \varphi_n$, that is, $\varphi \to \psi$.

Example 1.20 (Demonstration: the input)**.** A silly goal, since $p \to q$ was told, but a short run that shows every case. $\Gamma = \{p \to q\}$, the supposition is $\varphi = p$, and the given proof of q from $\Gamma \cup \{p\}$ is:

	$p \to q, p$	
1	$p \to q$	hypothesis: a member of Γ; case 1
2	p	hyptohesis: the supposition itself; case 2
3	q	MP 1, 2: case 3

The output will be a proof of $p \to q$ from Γ alone. Three lines in, one block of output for each.

[line 1, case 1] $\varphi_1 = p \to q$ is a member of Γ. Write it, weaken it by (I), fire.

	$p \to q$	
1	$p \to q$	hypothesis
2	$(p \to q) \to p \to (p \to q)$	(I) $[p \to q, p]$
3	$p \to (p \to q)$	MP 2, 1

Three output lines, ending in $p \to \varphi_1$.

[line 2, case 2] $varphi_2 = p$ is the supposition itself. Copy in the five line proof of $p \to p$.

[line 3, case 3] $varphi_3 = q$ acme by modus ponens from lines1 and 2. We hold $p \to (p \to q)$ at ouptut line 3 and $p \to p$. Distribute by (II), fire.

	$p \to q$	
1	$(p \to \to q) \to (p \to p) \to p \to q$	(II) $[p, p, q]$
2	$(p \to p) \to p \to q$	MP 1, $(p \to (p \to q))$
3	$p \to q$	MP 2, $(p \to p)$

1.5.1 The missing rule

To type a definition $f\,x := M$, a checker wants to say: suppose x has type A; if supposition M has type B , then f has type $A \to B$.

$$\Gamma, x : A \vdash M : B \implies \Gamma \vdash f : A \to B$$

With the programs rubbed out: if $\Gamma \cup \{\varphi\} \vdash \psi$ then $\Gamma \vdash \varphi \to \psi$. That *is* the Deduction Theoreom, left to right.

Example 1.21 (hypothetical syllogism, by supposing)**.** Suppose write access lets you publish, maintainers have accss, and Smith is a maintainer. Then, he may publish.

	$q \to r, p \to q, p$	
1	$q \to r$	hypothesis
2	$p \to q$	hypothesis
3	p	hypothesis
4	q	MP 2, 3
5	r	MP 1, 4

- Discharge $p : \{q \to r, p \to q\} \vdash p \to r$
- Discharge $p \to q : \{q \to r\} \vdash (p \to q) \to p \to r$.
- Discharge $q \to r :\vdash (q \to r) \to (p \to q) \to p \to r$

1.6 Summary

- A **proof** is a box of formulas, the hypotheses above the line, each line an axiom, a hypothesis, or modus ponens from earlier lines. Checking one is pattern matching: the clerk needs no understanding.
- A **typing is a proof** with the programs written beside the lines: **K** and **S** are the two schemas, application is modus ponens, a variable in the context is a hypothesis. The compiler that refused was refusing a bad proof.
- The rule the compiler needed for a parameter is the **Deduction Theorem**, $\Gamma \cup \{\varphi\} \vdash \psi$ iff $\Gamma \vdash \varphi \to \psi$, whose proof is the algorithm that compiles the paraemter away.
- When the language grew to **and, or and not**, seven schemas were added, each the type of a function we know, and neither theorem changed.